
생수다 BABP 3주차
" 정보로서 다뤄지는 Sequence"

How to Use GitHub



125jkop / SSD_BABP

Q Type / to search

<> Code

Issues

Pull requests

Actions

Projects

Wiki

Security and quality

Insights

Settings

Files

main

Go to file

01_Courses/A_Bioinformatics

Week01_Introduction_Setup

Week02_Weather_Analysis

Week03_Sequence_Database

.gitignore

LICENSE

README.md

SSD_BABP / 01_Courses / A_Bioinformatics

Add file

125jkop Add A_Week02 pdf and data files 7635b2c · 2 days ago History

Name	Last commit message	Last commit date
..		
Week01_Introduction_Setup	Add Week01 files	2 days ago
Week02_Weather_Analysis	Add A_Week02 pdf and data files	2 days ago
Week03_Sequence_Database	Add A_Week01 files	3 days ago

“Collect”

실험 디자인

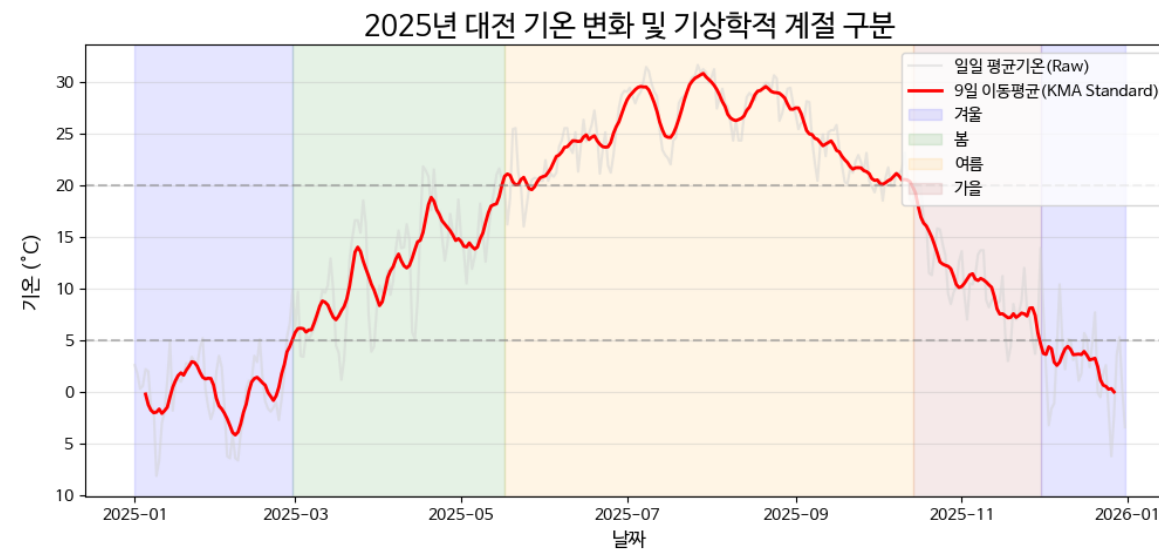
“Process”

필터링 & 정렬

“Analyze”

시각화 & 해석

Order of Data Analysis



“Collect”

실험 디자인

“Process”

필터링 & 정렬

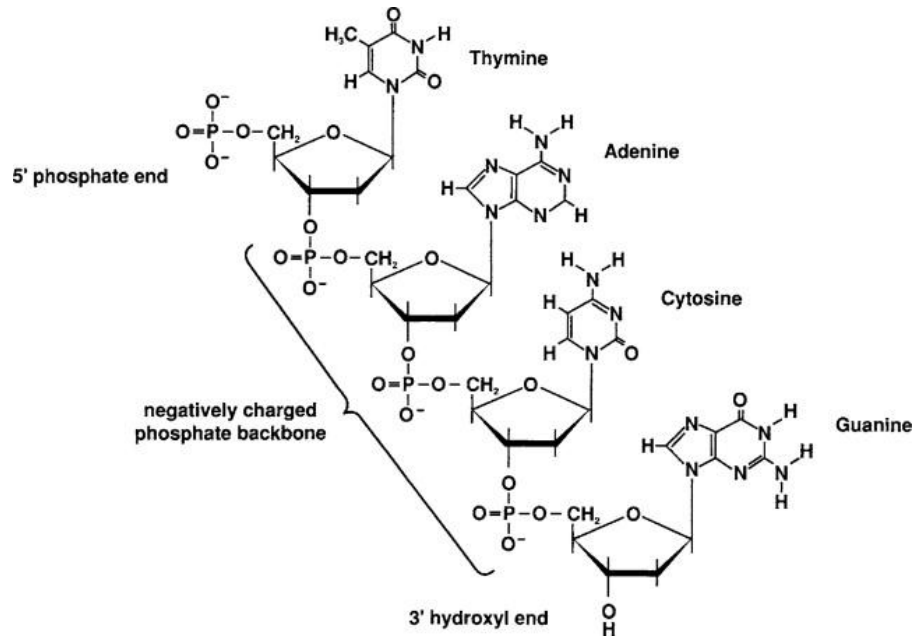
“Analyze”

시각화 & 해석

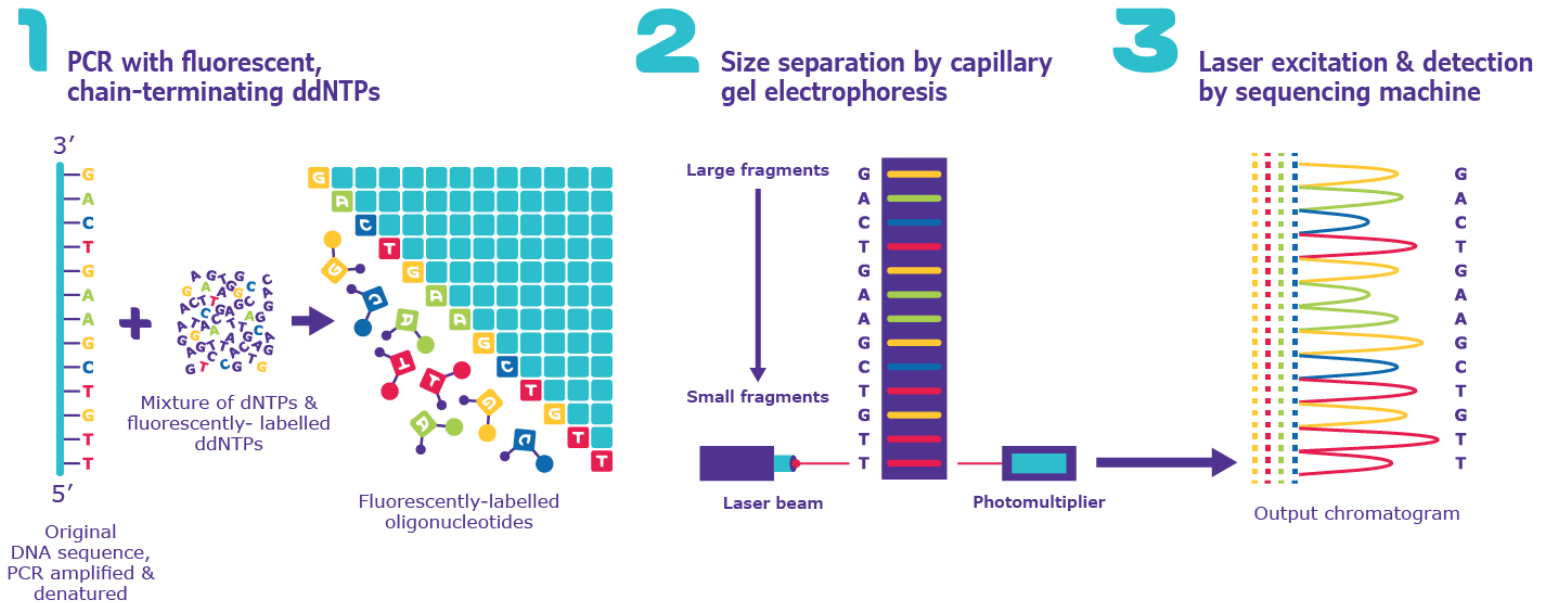
DNA Sequencing



Sanger sequencing

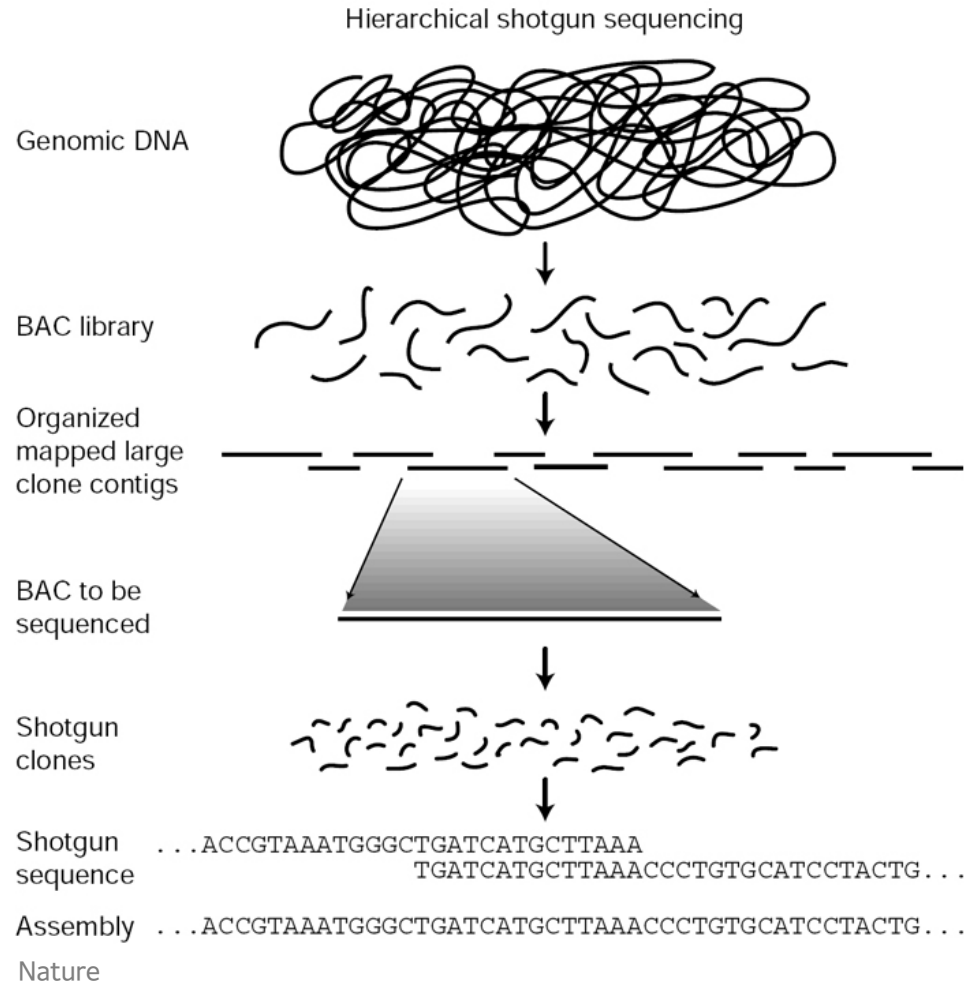


ScienceDirect



Sigma-Aldrich

Hierarchical shotgun sequencing

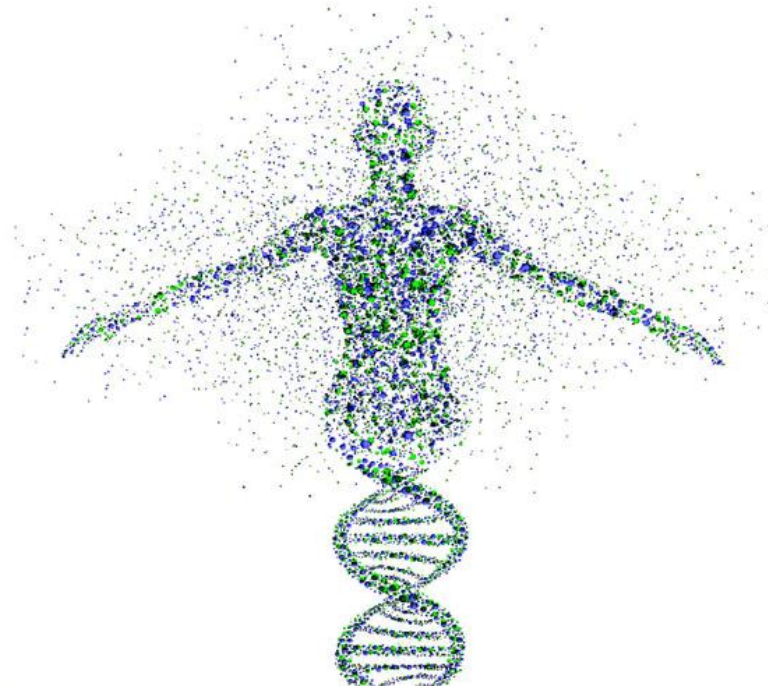


- 유전체를 큰 조각으로 나누어 BAC(Bacterial Artificial Chromosome)에 삽입
- STS(Sequence Tagged Site)로 상대적 위치를 결정하는 물리적 지도(Physical Map) 작성
- 클론들을 다시 무작위로 파쇄(Shotgun)하여 시퀀싱한 후, 미리 작성된 지도를 바탕으로 조립

The Human Genome Project



: 인간에게 존재하는 모든 염색체 내 염기 서열을 밝히고자 진행한 프로젝트



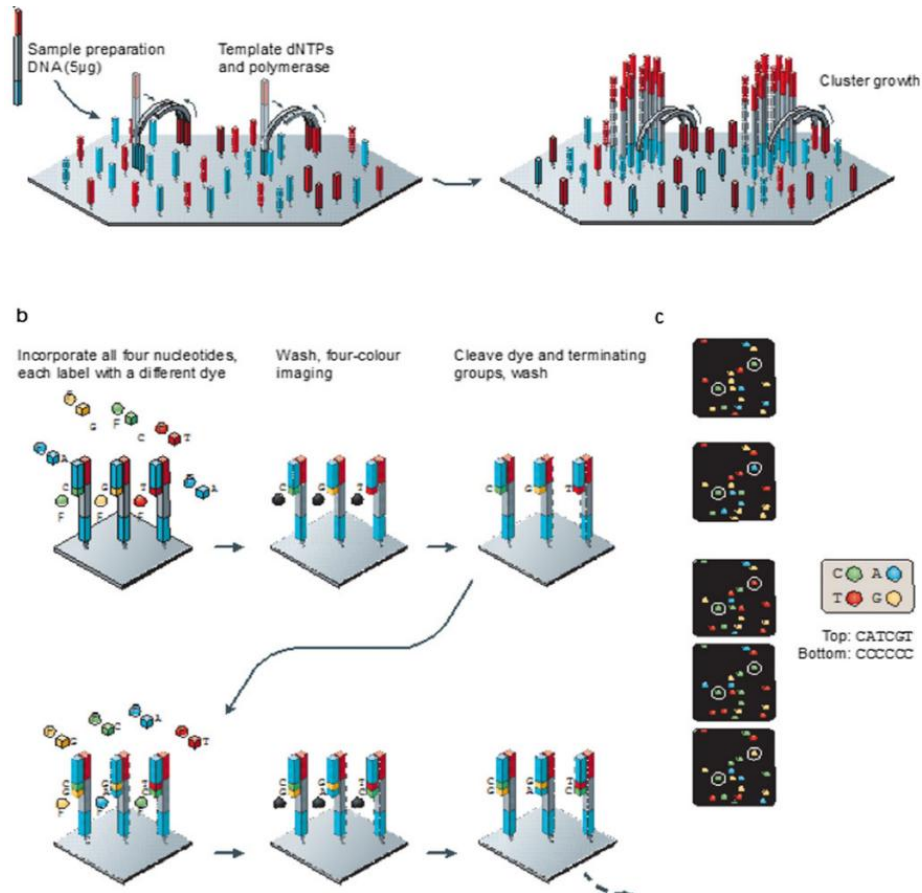
Amino Labs

의의: 인간의 생리적 기전 이해 증진, 정밀 의료 및 Reverse-genetics의 길을 개척
한계: 데이터의 편향, Centeromere, Telomere 등 반복 서열에 대한 부정확한 정보

Next Generation Sequencing



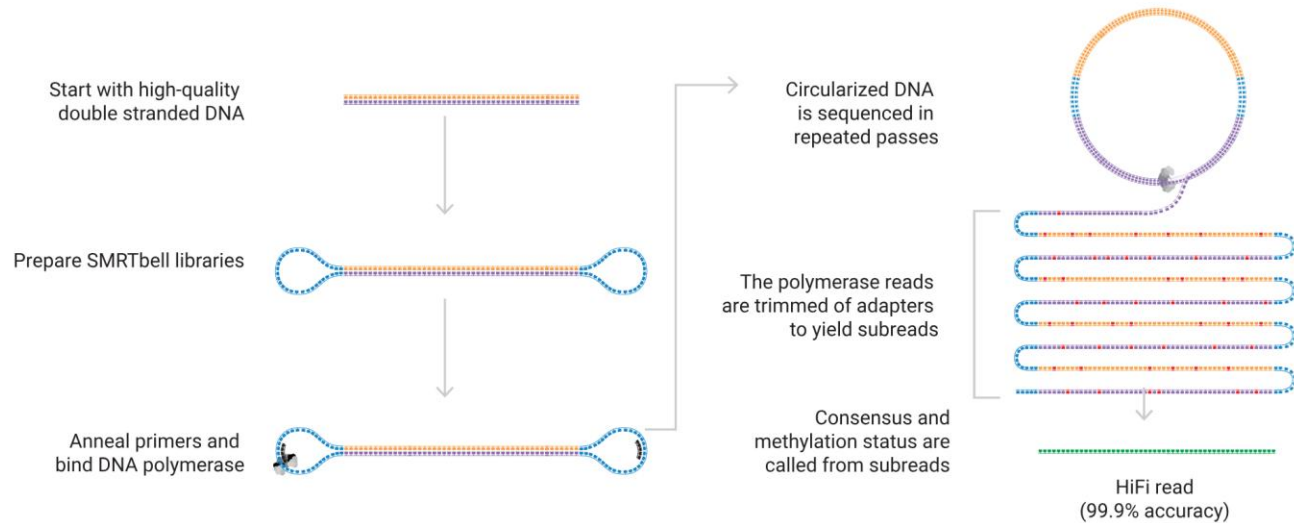
Illumina



Aydin S. (2016).

- Sample Prep: 서열에 Adapter와 Index 시퀀스를 부착하여 준비
- Flow cell: Adapter와 상보적인 서열이 존재하는 얇은 유리 판에 샘플을 흘려보내 부착
- Bridge Amplification: DNA polymerase가 각 샘플을 복제 한 후 단일 DNA 가닥이 구부러지며 근처 oligo와 재결합
→ 이를 반복해 하나의 read에 대하여 복제된 cluster를 얻음
- Signal Detection: 한 형광 물질 처리된 nucleotide가 DNA 가 결합할 때 발생하는 형광 신호를 측정하여 읽음

PacBio HiFi sequencing



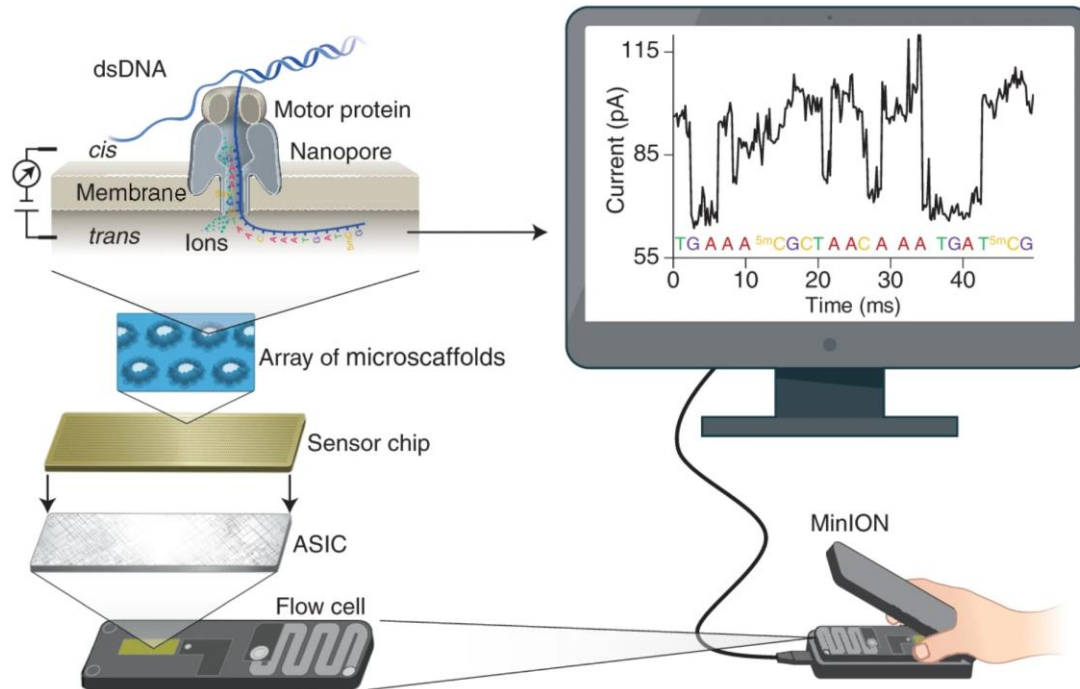
PacBio

- 분석할 긴 DNA 조각의 양 끝에 Adapter를 붙여 원형으로 제작
- 한 번에 긴 서열의 DNA loop를 Forward & Reverse로 번갈아가며 합성
- 합성할 때마다 실시간으로 형광 신호 감지

Next Generation Sequencing



Oxford Nanopore Technology



Wang, Y. *et al.* (2021).

- membrane에 nanopore를 만듦
- Pore 내로 DNA가 통과하면 각 염기마다 다른 크기로 전기 파동 신호를 포착

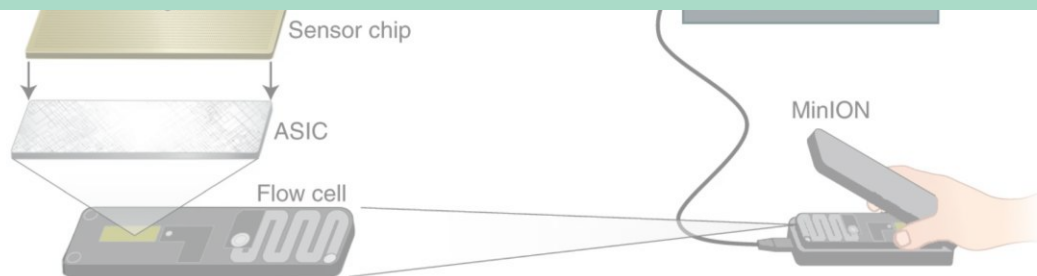
Next Generation Sequencing



Oxford Nanopore Technology



수집한 데이터는 어떻게 저장될까?



Wang, Y. *et al.* (2021).

나는 그기도 전기 파동 신호를 포식

The FASTA Format



FASTA ▾

Aequorea victoria green-fluorescent protein mRNA, complete cds

GenBank: M62653.1

[GenBank](#) [Graphics](#)

```
>M62653.1 Aequorea victoria green-fluorescent protein mRNA, complete cds
TACACACGAATAAAAGATAACAAAGATGAGTAAAGGAGAAGAAGCTTTTCACTGGAGTTGTCCCAATTCTT
GTTGAATTAGATGGTGATGTTAATGGGCACAAATTTTCTGTTCAGTGGAGAGGGTGAAGGTGATGCAACAT
ACGGAAAAGTTACCCCTAAATTTATTTGCACTACTGGAAAAGTACCTGTTCCATGGCCAACACTTGTCAC
TACTTTCTCTTATGGTGTTCAATGCTTTTCAAGATAACCCAGATCATATGAAACAGCATGACTTTTTCAAG
AGTGCCATGCCCGAAGGTTATGTACAGGAAAGAACTATATTTTTCAAAGATGACGGGAAGTACAAGACAC
GTGCTGAAGTCAAGTTTGAAGGTGATACCCCTTGTAAATAGAATCGAGTTAAAAGGTATTGATTTTAAAGA
AGATGGAAACATTCTTGGACACAAATTGGAATACAACATAACTCACACAATGTATACATCATGGCAGAC
AAACAAAAGAAATGGAATCAAAGTTAACTTCAAAATTAGACACAACATTGAAGATGGAAGCGTTCAACTAG
CAGACCATTATCAACAAAATACTCCAATTGGCGATGGCCCTGTCCTTTTACCAGACAACATTACCTGTC
CACACAATCTGCCCTTTCGAAAGATCCCAACGAAAAGAGAGAACCATGGTCCTTCTTGAGTTTGTAAAC
GCTGCTGGGATTACACATGGCATGGATGAACTATACAAATAAATGTCCAGACTTCCAATTGACACTAAAG
TGTCGAACAATTACTAAAATCTCAGGGTTCCTGGTTAAATTCAGGCTGAGATATTATTTATATATTTAT
AGATTCATTAATAATTGTATGAATAATTATTGATGTTATTGATAGAGGTTATTTCTTATTAACAGGCT
ACTTGGAGTGTATTCTTAATTCTATATTAATTACAATTTGATTTGACTTGCTCAA
```

Database for Bioinformatics: NCBI



 An official website of the United States government [Here's how you know](#) 

 **National Library of Medicine**
National Center for Biotechnology Information

Log in

All Databases 

Search

NCBI Home

Resource List (A-Z)

All Resources

Chemicals & Bioassays

Data & Software

DNA & RNA

Domains & Structures

Genes & Expression

Genetics & Medicine

Genomes & Maps

Homology

Literature

Proteins

Sequence Analysis

Taxonomy

Training & Tutorials

Variation

Welcome to NCBI

The National Center for Biotechnology Information advances science and health by providing access to biomedical and genomic information.

[About the NCBI](#) | [Mission](#) | [Organization](#) | [NCBI News & Blog](#) | [Visit the Beta Homepage](#)

Submit

Deposit data or manuscripts into NCBI databases



Download

Transfer NCBI data to your computer



Learn

Find help documents, attend a class or watch a tutorial



Develop

Use NCBI APIs and code libraries to build applications



Analyze

Identify an NCBI tool for your data analysis task



Research

Explore NCBI research and collaborative projects



Popular Resources

- [PubMed](#)
- [Bookshelf](#)
- [PubMed Central](#)
- [BLAST](#)
- [Nucleotide](#)
- [Genome](#)
- [SNP](#)
- [Gene](#)
- [Protein](#)
- [PubChem](#)

NCBI News & Blog

Updates to the NIH Manuscript Submission (NIHMS) System 05 May 2026

In Support of the 2024 NIH Public Access Policy As part of continued

Reminder! NCBI is Retiring rsync for FTP Downloads June 1, 2026 29 Apr 2026

Announcing brownout periods when rsync will be disabled As we previously

GenBank Release 271.0 is Available! 27 Apr 2026

GenBank release 271.0 (4/14/2026) is now available on the NCBI FTP site. This release has 53.90 trillion bases and 6.27

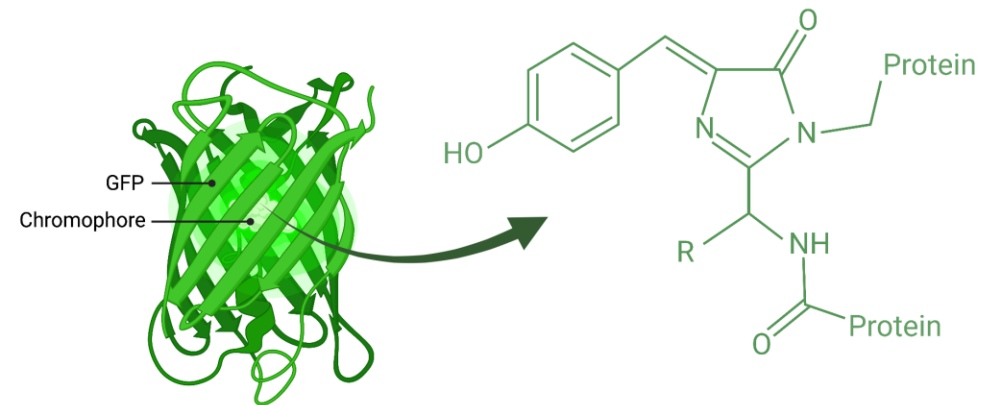
[More...](#)

Aequorea victoria



FineArtAmerica

Green Fluorescence Protein (GFP) with Chromophore



BioRender

“ β -barrel”

“Collect”

실험 디자인

“Process”

필터링 & 정렬

“Analyze”

시각화 & 해석



National Library of Medicine
National Center for Biotechnology Information

[Log in](#)

BLAST®

[Home](#) [Recent Results](#) [Saved Strategies](#) [Help](#)

Basic Local Alignment Search Tool

BLAST finds regions of similarity between biological sequences. The program compares nucleotide or protein sequences to sequence databases and calculates the statistical significance. [Learn more](#)

NEWS

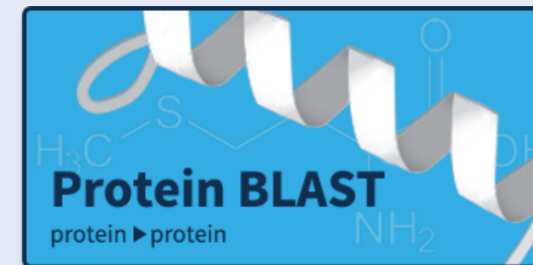
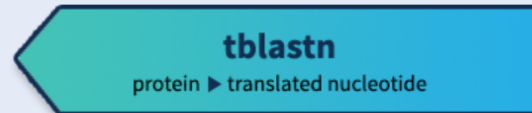
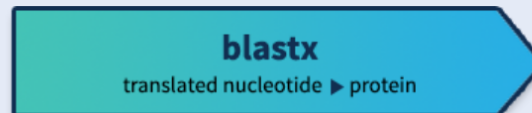
Mon, 21 Jul 2025

Here are a few highlights in our latest BLAST+ release:

Download BLAST+ 2.17.0 now!

[More BLAST news...](#)

Web BLAST



BLAST Genomes

Enter organism common name, scientific name, or tax id

Search

[Human](#)

[Mouse](#)

[Rat](#)

[Microbes](#)